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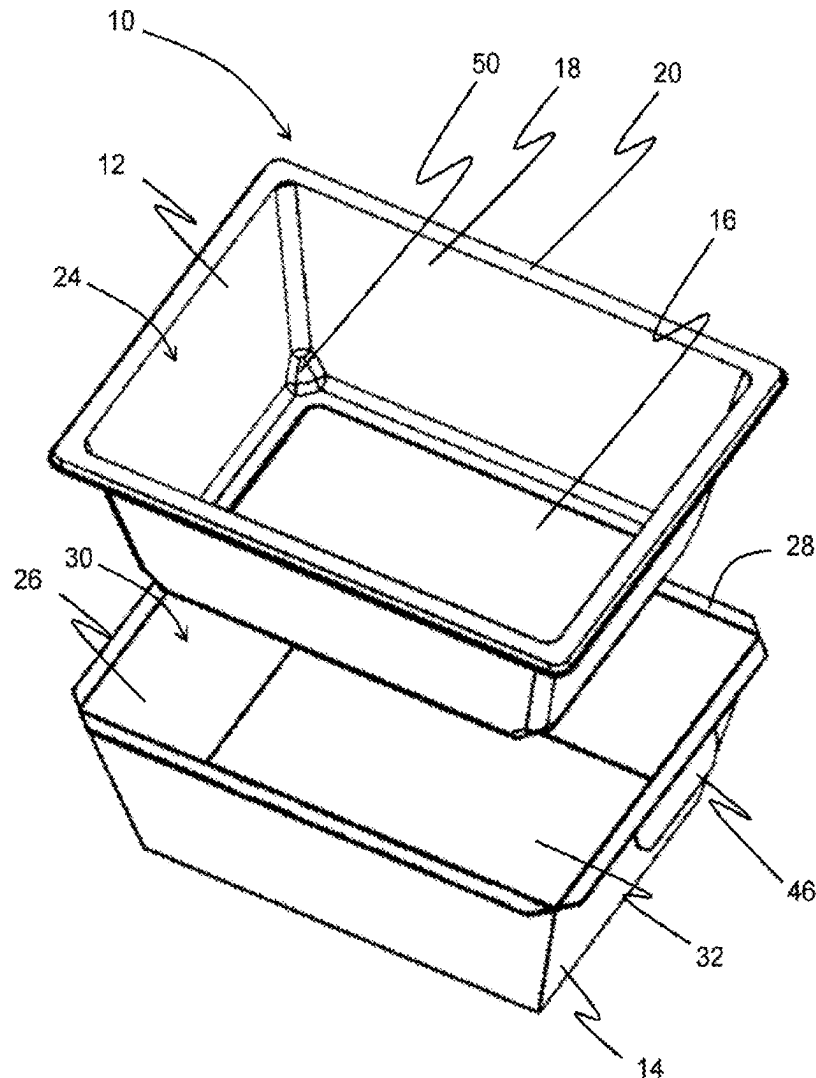
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ABSTRACT

A container assembly has a thermoformed plastic container with a base and a container wall terminating at a container flange, and a flexible container lid is attached to an upper surface of the container flange by a first heat sealing material. The assembly also includes a support with a support wall surrounding the container wall and terminating at a support flange, wherein the support flange is configured and arranged to abut against a lower surface of the container flange, and is attached thereto by a second heat sealing material.



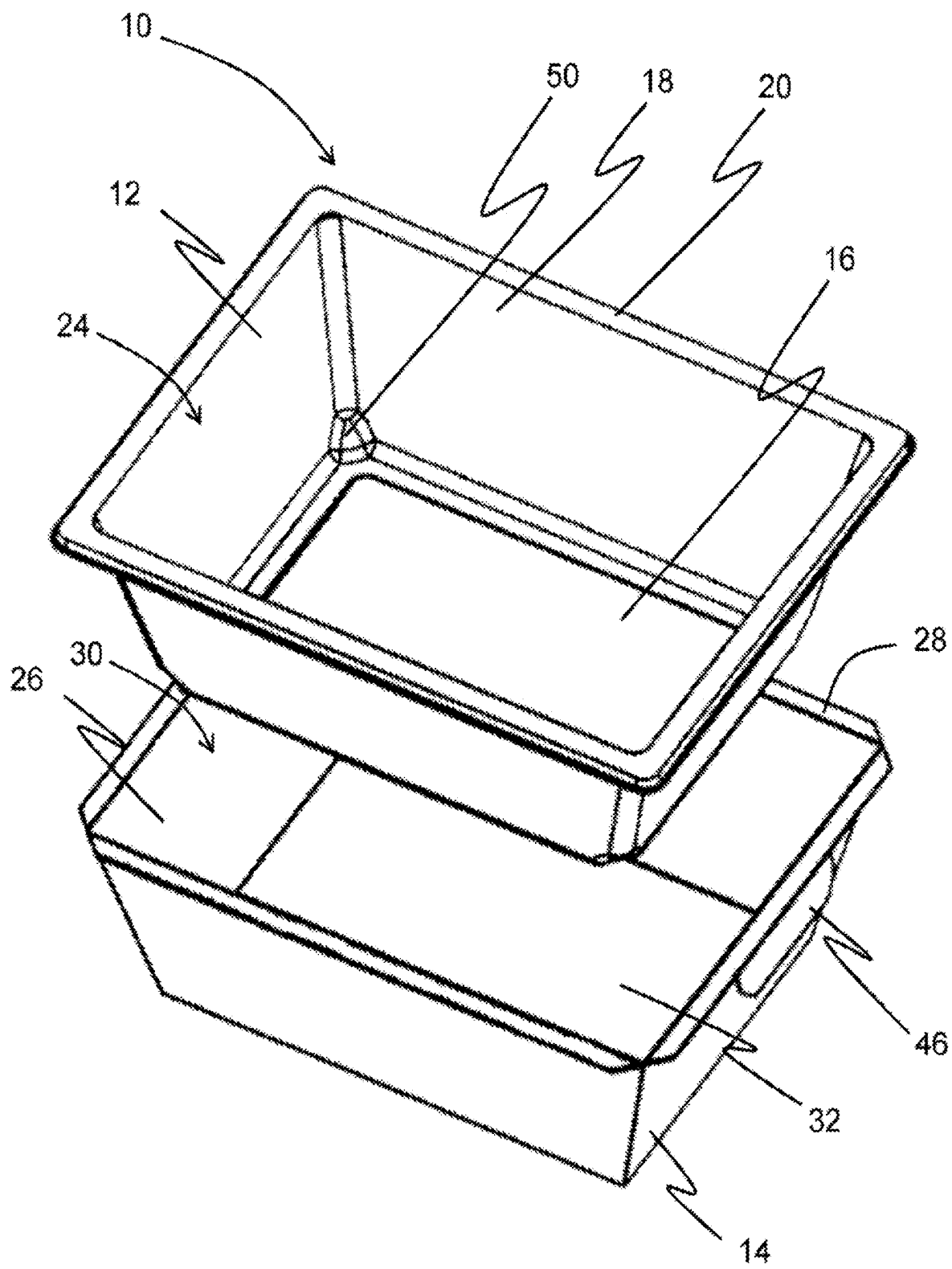


FIG. 1

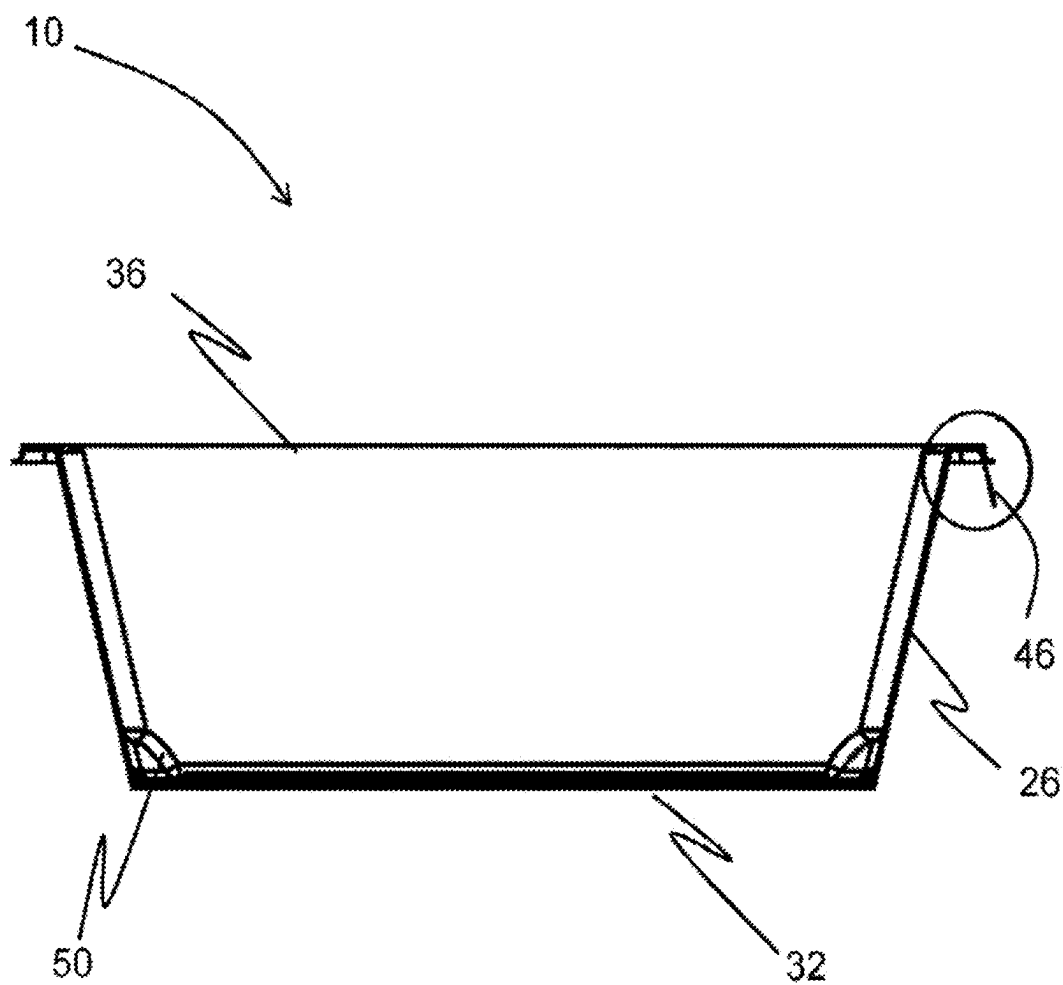
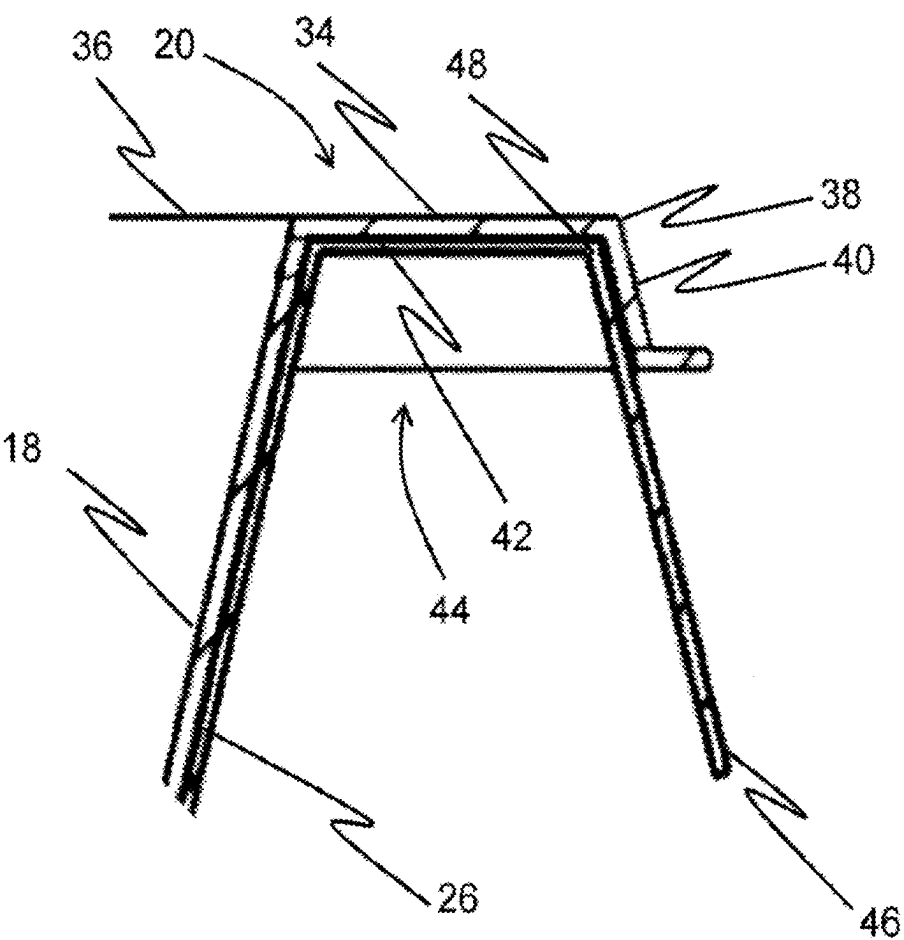


FIG. 2



14

FIG. 3

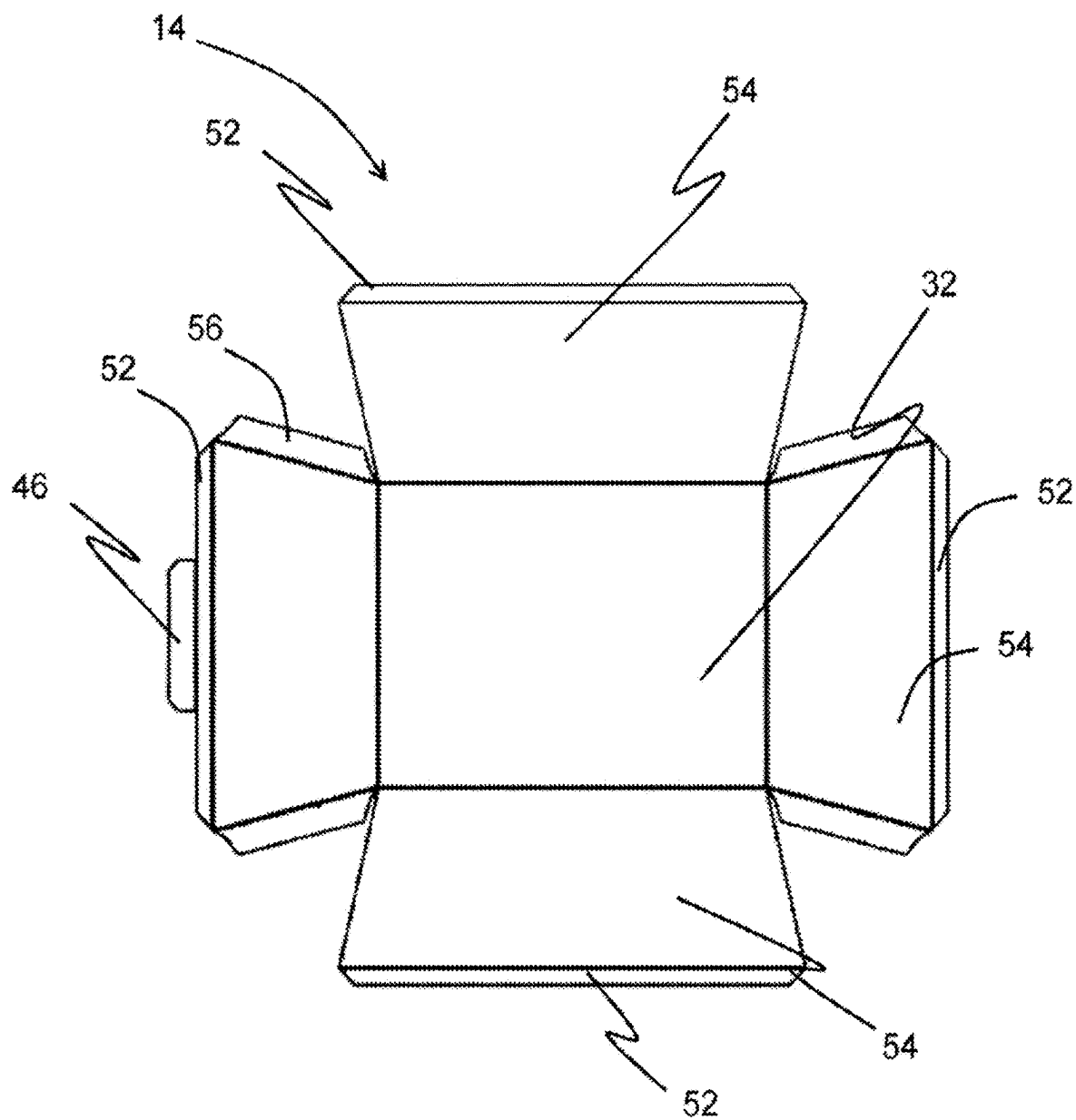


FIG. 4

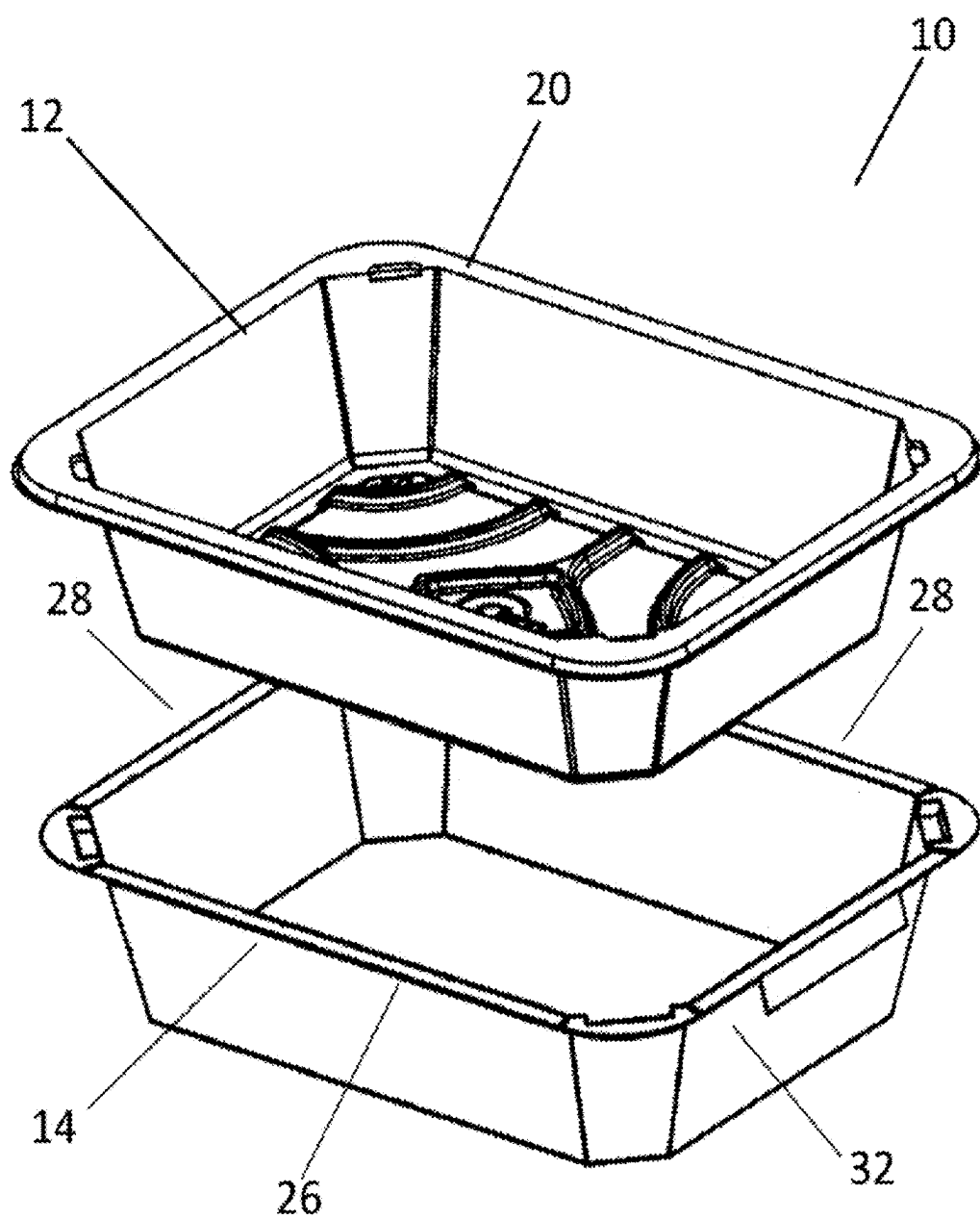


Figure 5

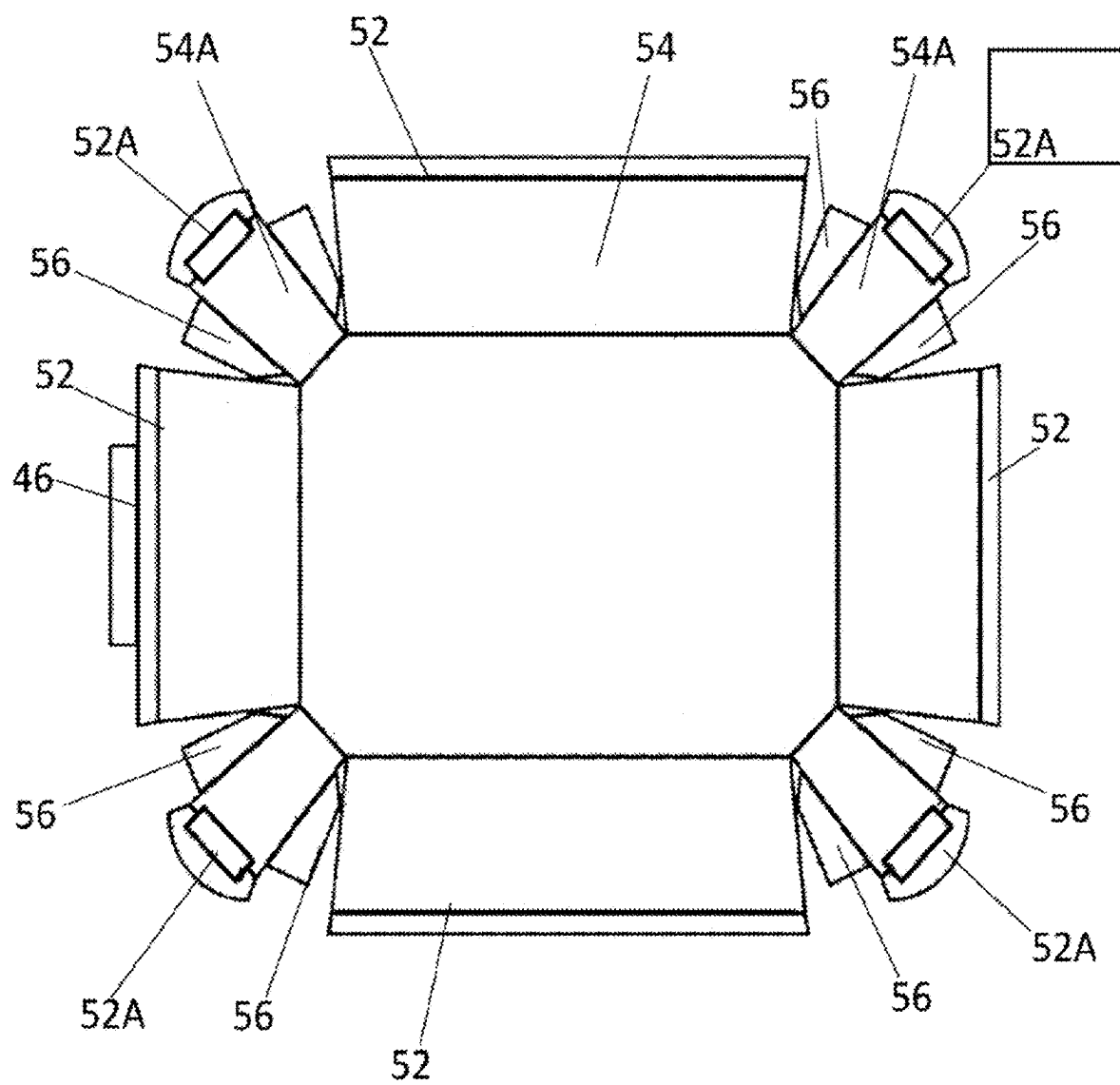


Figure 6

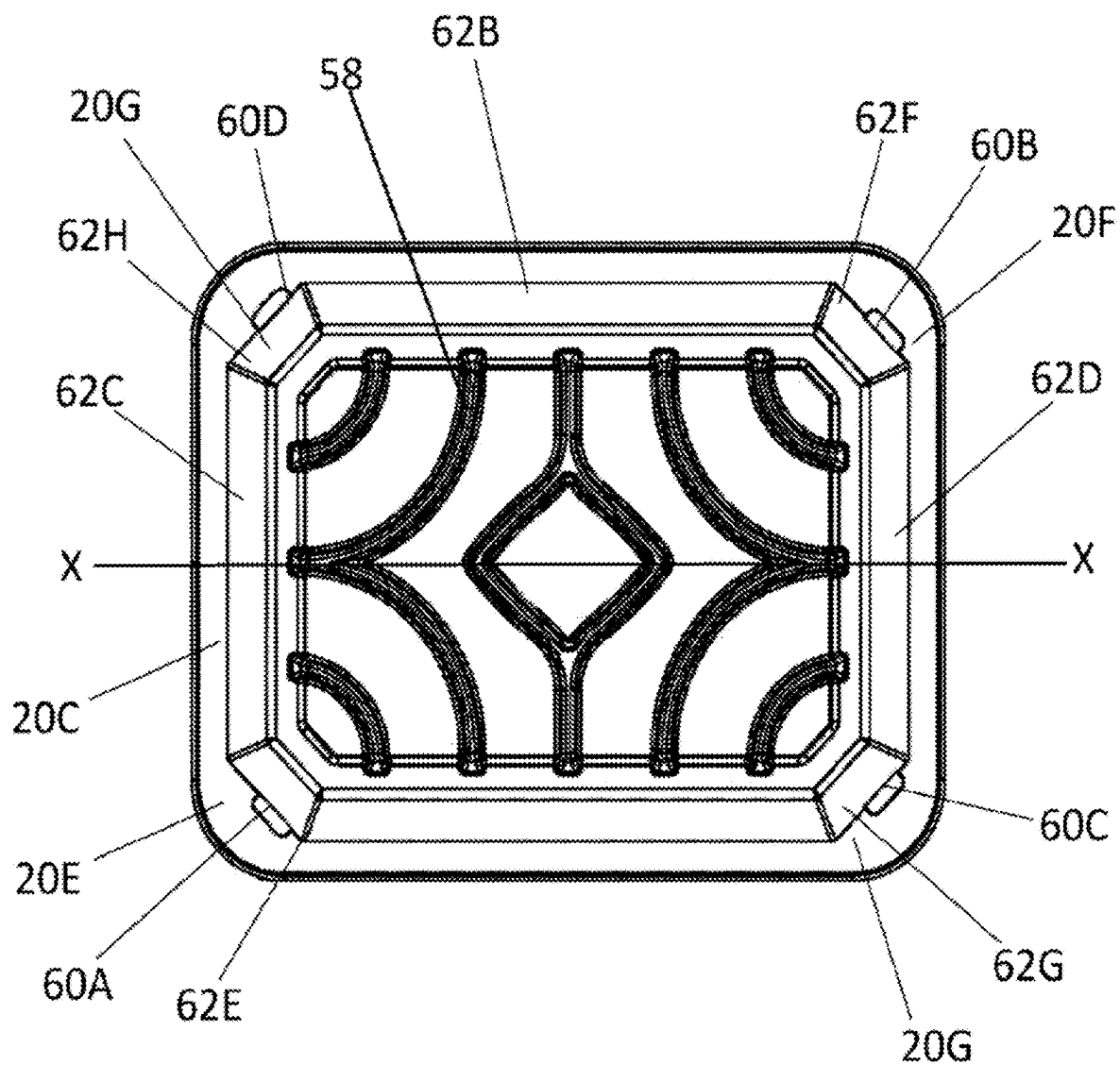


Figure 7

CONTAINER ASSEMBLY

FIELD

[0001] The present teachings relate to a container assembly, and to a method of forming a container assembly.

BACKGROUND

[0002] Thermoformed plastic containers are often used for packaging for a range of different product types and in a range of different sizes and shape. These thermoformed plastic containers are suitable for the storage and transportation of liquid or semi-solid foodstuffs such as, for example, soups, sauces, yoghurt and other dairy products, or solid foodstuffs such as, for example, nuts, dried fruit pieces and the like. These containers are also used to store and transport meat and fish, where modified atmosphere packaging (MAP) is often used to extend shelf life.

[0003] Despite rising awareness among consumers of global environmental factors such as global warming and rising marine pollution, the trend for consuming food and drinks from plastic containers is growing rapidly. Thus, the amount of plastics material, and so the raw materials such as oil, used to meet this demand is also increasing. This vast usage of raw materials results in a large carbon footprint for the plastics industry, thus contributing to global warming.

[0004] The present teachings seek to overcome or at least mitigate one or more problems associated with the prior art.

SUMMARY

[0005] According to a first aspect, there is provided a container assembly comprising: a thermoformed plastic container comprising a base, and a container wall extending from and surrounding the base and terminating at a container flange extending around the container wall to define a mouth of the container; a support comprising a support wall arranged to surround the container wall, said support wall terminating at a support flange; and a flexible container lid attached to an upper surface of the container flange by a first heat sealing material, wherein the support flange is configured and arranged to abut against a lower surface of the container flange, and wherein the support flange is attached to the lower surface of the container flange by a second heat sealing material.

[0006] The support provided by the support to the thermoformed plastic container enables a reduced amount of plastic material to be used to form the container. Attachment of the flexible container lid and the support to opposing upper and lower surfaces of the container flange enables the flexible container lid and the support to be secured to the container flange in a single step by applying heat and pressure to the top and bottom of the container flange. This facilitates the assembly process for the container assembly.

[0007] In some embodiments, heat may be applied to the upper surface of the container flange to heat the upper and lower surfaces of the container flange. Put another way, heat may be applied directly to the upper surface of the container flange and indirectly to the lower surface of the container flange through said container flange. In such embodiments, the melting temperature of the second heat seal material may be lower than that of the first heat sealing material.

[0008] In other embodiments, heat may be applied to the lower surface of the container flange to heat the upper and lower surfaces of the container flange. Put another way, heat

may be applied directly to the lower surface of the container flange and indirectly to the upper surface of the container flange through said container flange. In such embodiments, the melting temperature of the first heat seal material may be lower than that of the second heat sealing material.

[0009] In other embodiments, heat may be applied directly to the upper and lower surfaces of the container flange.

[0010] The support wall may abut against the container wall to provide support thereto. The tapered container wall may be seated on the tapered support wall. The container flange may be seated on the support flange.

[0011] The container flange may comprise a downwardly projecting lip extending from a peripheral edge of the container flange in a direction towards the base, and the lip may oppose the container wall to define a channel therebetween receiving the support flange therein. This arrangement has been found to facilitate assembly of the support onto the container as the channel acts as a locator for the support flange.

[0012] The support flange may be mechanically secured between the lip and the container wall. This arrangement facilitates assembly of the support and container prior to heat sealing the support and container by providing a mechanical attachment therebetween.

[0013] In some embodiments, an adhesive may be applied to the lower surface of the container flange and/or the upper surface of the support flange. The adhesive may be applied as one or more spots of adhesive on the container flange and/or support flange. The use of adhesive may help to hold the base of the container to the base of the support prior to heat sealing the support and container.

[0014] The support and the container may be configured and arranged to mechanically fix the support to the container. This arrangement has been found to facilitate assembly of the support and container prior to the attachment via the heat sealing materials.

[0015] The support may comprise a tab projecting from a distal edge of the support flange so as to extend beyond a distal end of the lip. This facilitates removal of the support from the container for separate recycling, after the container assembly has been used.

[0016] The container wall may comprise a plurality of container wall panels defining corner regions at an intersection between adjacent container wall panels and the base, and wherein each corner region is tapered or chamfered. This reduces the amount of plastic that is required to form the plastic container as it reduces the distance from a centre point of the container to the corners.

[0017] The support wall may comprise a plurality of support wall panels, and wherein the support flange comprises a support flange part extending from a distal edge of each support wall panel. The provision of the support flange being formed by multiple separate flange parts has been found to facilitate assembly of the support and container.

[0018] The support flange parts may be configured and arranged such that they form a continuous support flange around the support wall. Alternatively, the support flange parts may be configured and arranged such that they do not form a continuous support flange around the support wall.

[0019] Each support flange part may be tapered or chamfered at opposing ends thereof. This has been found to facilitate assembly of the support and container. In some

embodiments, each support flange part may be tapered or chamfered at an angle in the range 40°-50° at opposing ends thereof.

[0020] The container wall may be substantially linear in a direction between the base and the container flange and tapers in a direction towards the base. In some embodiments, the container wall may taper at an angle in the range 5° to 19°, in the range 10° to 18°, in the range 12° to 16°, for example approximately 14°. In some embodiments, the container wall may taper at an angle in the range 4° to 18°, in the range 8° to 16°, in the range 10° to 14°, for example approximately 12°. Providing angles in these ranges has been found to improve stacking of the containers side by side.

[0021] The support wall may conform to the shape of the container wall. This helps to improve the support provided by the support to the container.

[0022] The flexible lid may be formed from a plastic film, a metal foil, a paper material, and/or a card material. In some embodiments the flexible lid may be coated with the first heat sealing material.

[0023] The first heat sealing material may be selected from: a non-plastic heat sealable material such as polylactic acid; an aqueous heat-sealing material; and/or a polymeric film such as polyethylene, polypropylene, polyethylene terephthalate.

[0024] In some embodiments, the second heat sealing material may be selected from: a non-plastic heat sealable material such as polylactic acid; an aqueous heat-sealing material; and/or a polymeric film such as polyethylene, polypropylene, and/or polyethylene terephthalate.

[0025] The support may be formed from a fibrous material such as paper or card.

[0026] In some embodiments, the support may be formed from a folded and glued blank, formed for example from paper or card. In alternatively embodiments, the support may be a pre-formed pulp board container (like high quality egg boxes).

[0027] The support flange may be coated with the second heat sealing material.

[0028] The support wall may comprise a plurality of support wall panels defining corner regions at an intersection between adjacent container wall panels. The support flange may extend continuously around support wall or may not extend continuously around the support wall. In either case, the support flange may extend around, or into, the corners of the support wall. Put another way, the corner regions of the support wall may terminate at the support flange.

[0029] The container wall may comprise eight container wall panels. The support wall may also comprise eight support wall panels.

[0030] According to a second aspect, there is provided a method of forming a container assembly, the method comprising the steps of: thermoforming a plastic container comprising a base, and a container wall extending from and surrounding the base and terminating at a container flange extending around the container wall to define a mouth of the container; providing a support comprising a support wall terminating at a support flange; positioning the container within the support such that the support wall surrounds the container wall and the support flange abuts against a lower surface of the container flange; positioning a flexible container lid over an upper surface of the container flange; and applying heat and pressure to the upper and lower surfaces

of the container flange to attach the flexible container lid to the upper surface of the container flange and the support flange to the lower surface of the container flange via first and second heat sealing materials, respectively.

[0031] The step of applying heat and pressure to the upper and lower surfaces of the container flange may comprise simultaneously applying heat and pressure to the upper and lower surfaces of the container flange.

[0032] In some embodiments, heat may be applied to the upper surface of the container flange to heat the upper and lower surfaces of the container flange. Put another way, heat may be applied directly to the upper surface of the container flange and indirectly to the lower surface of the container flange through said container flange. In such embodiments, the melting temperature of the second heat seal material may be lower than that of the first heat sealing material.

[0033] In other embodiments, heat may be applied to the lower surface of the container flange to heat the upper and lower surfaces of the container flange. Put another way, heat may be applied directly to the lower surface of the container flange and indirectly to the upper surface of the container flange through said container flange. In such embodiments, the melting temperature of the first heat seal material may be lower than that of the second heat sealing material.

[0034] In other embodiments, heat may be applied directly to the upper and lower surfaces of the container flange.

[0035] The method may comprise the step of forming the flexible container lid from the first heat seal material.

[0036] The method may comprise the step of coating the flexible container lid and/or the upper surface of the container flange with the first heat seal material.

[0037] The method may comprise the step of coating the lower surface of the container flange and/or the support flange with the second heat seal material.

BRIEF DESCRIPTION OF THE DRAWINGS

[0038] Embodiments will now be described with reference to the accompanying drawings, in which:

[0039] FIG. 1 is an exploded isometric view of a container assembly according to an embodiment;

[0040] FIG. 2 is a cross-sectional side view of the container assembly of FIG. 1;

[0041] FIG. 3 is an enlarged cross-sectional side view of the circled region of FIG. 2;

[0042] FIG. 4 is a plan view of the support of the container assembly of FIG. 1, prior to assembly;

[0043] FIG. 5 is an exploded isometric view of a container assembly according to an embodiment;

[0044] FIG. 6 is a plan view of the support of the container assembly of FIG. 5, prior to assembly; and

[0045] FIG. 7 is a plan view of the container of the container assembly of FIG. 5.

DETAILED DESCRIPTION OF EMBODIMENT(S)

[0046] Referring to FIGS. 1 to 3, there is shown a container assembly generally designated 10. The container assembly 10 includes a plastic container 12 and a support 14. The container 12 may be a thermoformed plastic container. The container 12 is positioned in and is supported by the support 14. The support provided to the thermoformed plastic container 12 enables a reduced amount of plastic material to be used to form the container 12.

[0047] The container 12 has a base 16 and a container wall 18 surrounding the base 16. The container wall 18 extends from the base 16. The container wall 18 extends to and terminates at a container flange 20 extending around the periphery of the container 12. Put another way, the container flange 20 extends around the periphery of the container wall 18. The container flange 20 extends around an entirety of the periphery of the container 12 (i.e. around an entirety of the container wall 18). The base 16 and container wall 18 co-operate to define an interior space of the container 12. The container flange 20 defines a mouth 24 of the container 12.

[0048] The container 12 may be manufactured from sheet polyethylene terephthalate (PET). The container 12 may alternatively be manufactured from sheet polypropylene (PP), from crystallised polyester (CPET), or from amorphous polyethylene terephthalate (APET). It will be understood that the term “thermoforming” refers to the forming of the container by the thermal softening of a sheet of plastic material, which is then conformed to the shape of a mould. In alternative arrangements, any suitable plastics material may be used. It will be appreciated that the sheet material may have a thickness of less than 300 microns, for example in the range 150 to 250 microns, for example approximately 200 microns. Providing a sheet of these thicknesses have been found to ensure that the thickness of the resulting container is sufficient (e.g. at least 100 microns) over the entire container.

[0049] As an alternative, the container 12 may be formed from sheet RPET that can contain up to 100% recycled post-consumer waste. In yet a further embodiment, the container 10 may be formed from laminated PET/RPET having an A/B/A structure with 100% RPET in the central layer and two outer layers of PET. Any thickness combination is possible. For a sheet thickness of 1000 micron, the layers may be as follows—with 100 PET/800 RPET/100 PET.

[0050] The support 14 is fitted to the container 12 to form the container assembly 10. The support 14 includes a support wall 26 that surrounds the container wall 18. Put another way, the support wall 26 defines an opening into which the container 12 is positioned. The support wall 26 terminates at a support flange 28. The support flange 28 defines a mouth 30 or upper opening of the support 14. In the illustrated arrangement, the support 14 includes a base 32 such that the support 14 may be considered to be a tray. In alternative arrangements, the base 32 may be omitted and the support 14 may be considered to be a sleeve. The support 14 may be formed from a fibrous material such as paper or card. The support 14 (i.e. the tray or sleeve) may be provided as a decorative card. The card which forms the support 14 may have a thickness of less than 650 microns, for example less than 500 microns, for example, less than 400 microns, for example in the range 250 to 400 microns, for example in the range 300 to 350 micron, for example in the range 330 to 340 microns, for example approximately 335 microns.

[0051] The container flange 20 includes a substantially flat or planar upper surface 34 that surrounds the mouth 24 of the container 12. The upper surface 34 defines an attachment location for a flexible container lid 36. The plane of the upper surface 34 is substantially parallel to the plane of the base 16 of the container 12. The container flange 20 includes a peripheral edge 38. The peripheral edge 38 is defined by a lip 40 extending in a direction towards the base 16. The

peripheral edge 38 of the container flange 20 may be used to retain a rigid replaceable plastic lid (not shown) that is configured to fit over the container flange 20.

[0052] The flexible container lid 36 may be formed from a plastic film, a metal foil, paper or card. The flexible container lid 36 is attached to the upper surface 34 of the container flange 20 by a first heat sealing material (not shown). The first heat sealing material may be selected from a polymeric film such as polyethylene, polypropylene, polyethylene terephthalate, and/or a metal foil such as aluminium foil. It will be appreciated that the flexible container lid may be formed from the first heat sealing material, or may be coated with the first heat sealing material. Alternatively, it will be understood that the container flange may be coated with the first heat sealing material. The flexible container lid 36 is attached to the upper surface 34 by the application of heat and pressure.

[0053] The container flange 20 includes a substantially flat or planar lower surface 42. The lower surface 42 defines an underside of the container flange 20. The support flange 28 is configured and arranged to abut against the lower surface 42 of the container flange 20, as is shown in FIG. 3. Put another way, the container flange is seated on the support flange. The support flange 28 is attached to the lower surface 42 of the container flange 20 by a second heat sealing material (not shown). The second heat sealing material may be a non-plastic heat sealable material, such as polylactic acid. In some arrangements, the second heat sealing material may be an aqueous heat sealing material, for example aqueous dispersions of low-density polyethylene and/or ethylene vinyl acetate. In some arrangements, the second heat sealing material may be polymeric film such polyethylene, polypropylene, polyethylene terephthalate. The second heat sealing material may be the same or different to the first heat sealing material. It will be appreciated that the support flange 28 may be coated with the second heat sealing material, or the lower surface 42 of the container flange 20 may be coated with the second heat sealing material. The support 14, i.e. the support flange 28, is attached to the lower surface 42 by the application of heat and pressure.

[0054] The flexible container lid 36 and the support flange 28 are attached to opposing upper and lower surfaces 34, 42 of the container flange 20 via heat sealing materials. This arrangement enables the flexible container lid 36 and the support flange 28 to be secured to the container flange 20 in a single step by applying heat and pressure to the upper and lower surfaces 34, 42 of the container flange 20.

[0055] In some embodiments, heat may be applied to the upper surface of the container flange to heat the upper and lower surfaces of the container flange. Put another way, heat may be applied directly to the upper surface of the container flange and indirectly to the lower surface of the container flange through said container flange. In such embodiments, the melting temperature of the second heat seal material may be lower than that of the first heat sealing material. In other embodiments, heat may be applied to the lower surface of the container flange to heat the upper and lower surfaces of the container flange. Put another way, heat may be applied directly to the lower surface of the container flange and indirectly to the upper surface of the container flange through said container flange. In such embodiments, the melting temperature of the first heat seal material may be lower than that of the second heat sealing material. In other

embodiments, heat may be applied directly to the upper and lower surfaces of the container flange.

[0056] As discussed above, the container flange 20 includes a lip 40 extending from a peripheral edge 38 of the container flange 20 in a direction towards the base 16. The support 14 a tab 46 projecting from a distal edge 48 of the support flange 28. The tab 46 is provided to enable a user to separate the support 14 from the container 12 after the container assembly 10 has been used. The tab 46 is arranged so as to extend beyond a distal end 48 of the lip 40.

[0057] The lip 40 is arranged to oppose the container wall 18 to define a channel 44 therebetween. The channel 44 received the support flange 28 therein. The support flange 28 may be mechanically secured to the container 12. Put another way, the support flange 28 may be mechanically secured between the lip 40 and the container wall 18. The support flange 28 may be dimensioned such that it abuts against the lip 40 and the container wall 18 to mechanically secure the support 14 to the container 12. In some embodiments, an adhesive may be applied, e.g. as one or spots, to the lower surface of the container flange and/or the upper surface of the support flange to hold the base of the container to the base of the support prior to heat sealing the support and container.

[0058] In the illustrated arrangement, the container 12 and the support 14 are generally rectangular in plan-view. The container 12 and support 14 each include four container wall panels or side walls. The wall panels are substantially planar and are arranged to form the container wall 18 and support wall 26, respectively. In alternative arrangements, it will be appreciated that the container 12 and support 14 may be any suitable shape in plan-view such as circular, triangular, rectangular, pentagonal, hexagonal, octagonal or any other suitable shape. It will be appreciated that the container 12 may have other shapes and volumes to suit the application.

[0059] The container wall 18 is substantially linear in a direction between the base 16 and the container flange 20. The container wall 18 tapers in a direction towards the base 16. Put another way, the container wall 18 defines a width that reduces from a maximum width at the container flange 20 to a minimum width at the base 16. This tapered wall configuration enables empty containers 12 to be nested within one another for storage and transportation. The container wall 18 tapers at an angle in the range 5° to 19°, in the range 10° to 18°, in the range 12° to 16°, for example approximately 14°. Alternatively, the container wall 18 may taper at an angle in the range 4° to 18°, in the range 8° to 16°, in the range 10° to 14°, for example approximately 12°.

[0060] The shape of the support wall 26 conforms to the shape of the container wall 18. Put another way, the support wall 26 is substantially linear and tapers in a direction away from the support flange 28. In this way, the support wall 26 abuts against the container wall 18 to provide support thereto. Put another way, the tapered container wall 18 is seated on the tapered support wall 26. It will be appreciated that the features of the container wall 18 described above are also applicable to the support wall 26. Providing a support wall 26 that conforms to the shape of the container wall 18 has been found to improve the support provided to the container 12.

[0061] The container wall 18 is formed from four container side walls or container wall panels extending around the base 16. In this way, the container 12 defines corner regions 50 at the intersection between adjacent container

side walls and the base 16. Each corner region 50 is tapered or chamfered. This arrangement reduces the distance from a centre point of the container 12 to the corner regions 50 of the container 12, which reduces the thickness of plastic material needed to thermoform the container 12.

[0062] Referring now to FIG. 4, the support 14 is provided in the form of a foldable blank that is to be assembled prior to fitting to the container 12. In this arrangement, the support wall 26 is formed from four support wall panels or support side wall panels 54. It will be appreciated that in alternative arrangements any suitable number of wall panels 54 may be provided. Each wall panel 54 is connected to and extends from the base 32, but in arrangements where the base 32 is omitted it will be understood that one or more of the wall panels 54 may be foldably connected.

[0063] The support may be provided with one or more attachment flaps 56 to secure adjacent support wall panels together to form the support wall 26. Attachment may be achieved via an adhesive or any other suitable means. In the illustrated embodiment, the end wall panels are each provided with two attachment flaps to secure the two adjacent side wall panels thereto. It will be understood, however, that any suitable arrangement of the attachment flaps may be used to secure the support wall panels together to form the support wall 26. In the present arrangement, one or more of the wall panels 54 are provided with an attachment flap 56 extending from a lateral edge of the wall panel 54 and arranged to contact an adjacent wall panel 54 during assembly of the support 14. It will be understood that the attachment flap(s) 56 will be attached to the adjacent wall panel 54 via the adhesive, which may be applied or coated on the attachment flap 56 and/or the adjacent wall panel 54.

[0064] The support flange 28 is provided as a support flange part 52 extending from a distal edge of each support side wall panel 54. The provision of the support flange 28 being formed from multiple separate flange parts has been found to facilitate assembly of the support 14 and container 12. In the illustrated arrangement, the support flange parts 52 are configured and arranged such that they do not form a continuous flange around the support wall 26. In the arrangement shown, each support flange part 52 is tapered at opposing ends thereof. The taper at each end of the support flange parts 52 may be at an angle in the range 30°-60°, for example in the range 40°-50°, for example approximately 45°. In some alternative arrangements, each flange support part 52 may not be tapered and may extend substantially perpendicularly from the distal edge of the respective side wall panel 54. In further alternative arrangements, the support flange parts 52 may be configured and arranged to form a continuous flange around the support wall 26.

[0065] Referring now to FIGS. 5 to 7, a second embodiment of the container assembly 10 is shown. Like the container assembly 10 of the first embodiment, the container assembly 10 of the second embodiment comprises a container 12 and a support 14. The container 12 may be a thermoformed plastic container, as has been described with reference to FIGS. 1 to 3. The container 12 is positioned in and supported by the support 14. The support 14 may be formed from a fibrous material such as paper or card. The second embodiment has many elements in common with the container assembly of FIGS. 1 to 4, and hence, in the following, only differences will be described in detail.

[0066] The container 12 has a base 16 and a container wall 18 surrounding the base 16. The container wall 18 extends

from the base 16. The container wall 18 extends to and terminates at a container flange 20 extending around the periphery of the container 12. Put another way, the container flange 20 extends around the periphery of the container wall 18. The container flange 20 extends around an entirety of the periphery of the container 12 (i.e. around an entirety of the container wall 18). The base 16 and container wall 18 co-operate to define an interior space of the container 12. The container flange 20 defines a mouth 24 of the container 12.

[0067] The container flange 20 includes a lip 40 extending from a peripheral edge 38 of the container flange 20 in a direction towards the base 16. The support 14 a tab 46 projecting from a distal edge 48 of the support flange 28. The tab 46 is provided to enable a user to separate the support 14 from the container 12 after the container assembly 10 has been used. The tab 46 is arranged so as to extend beyond a distal end 48 of the lip 40.

[0068] The lip 40 is arranged to oppose the container wall 18 to define a channel 44 therebetween. The channel 44 received the support flange 28 therein. The support flange 28 may be mechanically secured to the container 12. Put another way, the support flange 28 may be mechanically secured between the lip 40 and the container wall 18. The support flange 28 may be dimensioned such that it abuts against the lip 40 and the container wall 18 to mechanically secure the support 14 to the container 12. In some embodiments, an adhesive may be applied, e.g. as one or spots, to the lower surface of the container flange and/or the upper surface of the support flange to hold the base of the container to the base of the support prior to heat sealing the support and container.

[0069] The container 12 of the second embodiment includes eight wall panels. These include a first pair of equal and opposite lengthwise wall panels which extend parallel to the length of the container 12. Put another way, the container includes a first pair of side wall panels. The container 12 includes a second pair of equal and opposite width-wise wall panels which extend parallel to the width of the container 12. Put another way, the container 12 include a second pair of end wall panels. The container 12 includes four corner wall panels which extend between and connect the lengthwise and widthwise wall panels to form a continuous container wall. Put another way, each corner container wall panel extending between a side wall panel and an adjacent end wall panel. The side wall panels define a substantially octagonal container. The side wall panels are arranged perpendicular to the end wall panels, and each corner wall panel extends at an angle between the adjacent end wall and side wall of the container. Each of the corner walls panels spans a smaller distance than either the lengthwise or widthwise wall panels, such that the container 12 substantially describes a rectangle with chamfered corners in plan-view. Similar to the container of FIGS. 1 to 3, the container wall 18 extends to and terminates at a container flange 20 which extends around the periphery of the container 12 and which defines a mouth 24 of the container 12. The container flange defines a perimeter that is substantially rectangular, for example the perimeter defines a rectangle with curved corners.

[0070] The support 14 of the second embodiment includes eight wall panels. These include a first pair of equal and opposite lengthwise wall panels which extend parallel to the length of the support 14. Put another way, the support

includes a first pair of side support wall panels. The support 14 includes a second pair of equal and opposite width-wise wall panels which extend parallel to the width of the support 14. Put another way, the support 14 includes a second pair of end support wall panels. The support 14 includes four corner wall panels which extend between and connect the lengthwise and widthwise wall panels to form a continuous support wall. Put another way, the support includes four corner support wall panels each corner support wall panels extending a support side wall panel and an adjacent support end wall panel. The support side wall define a substantially octagonal support 14. The side wall panels are arranged perpendicular to the end wall panels, and each corner wall panel extends at an angle between the adjacent end wall and side wall of the container. Each of the corner walls panels spans a smaller distance than either the lengthwise or widthwise wall panels, such that the support 14 substantially describes a rectangle with chamfered corners in plan-view. Similar to the support 14 of FIGS. 1 to 4, the support wall 26 terminates at a support flange 28 which defines a mouth 30 or upper opening of the support 14. The support flange defines a perimeter that is substantially rectangular, for example the perimeter defines a rectangle with curved corners.

[0071] As described above in connection with the container assembly of FIGS. 1 to 4, the support flange 28 is attached to the lower surface 42 of the container flange 20 by a second heat sealing material (not shown). The second heat sealing material may be a non-plastic heat sealable material, such as polylactic acid. In some arrangements, the second heat sealing material may be an aqueous heat-sealing material, for example aqueous dispersions of low-density polyethylene and/or ethylene vinyl acetate. In some arrangements, the second heat sealing material may be polymeric film such polyethylene, polypropylene, polyethylene terephthalate. The second heat sealing material may be the same or different to the first heat sealing material. It will be appreciated that the support flange 28 may be coated with the second heat sealing material, or the lower surface 42 of the container flange 20 may be coated with the second heat sealing material. The support 14, i.e. the support flange 28, is attached to the lower surface 42 by the application of heat and pressure.

[0072] Referring now to FIG. 6, the foldable blank used to form the support 14 of the second embodiment includes eight support wall panels or support side wall panels 54. The support side wall panels 54 include four corner wall panels 54A which define the chamfered corners of the rectangle described by the support 14 in plan-view when the foldable blank is assembled into the support 14. In the arrangement shown, the attachment flaps 56 which join the support wall panels 54 to form a continuous support wall are provided on the lateral edges of the corner wall panels 54A.

[0073] The support may be provided with one or more attachment flaps 56 to secure adjacent support wall panels together to form the support wall 26. Attachment may be achieved via an adhesive or any other suitable means. In the illustrated embodiment, the corner wall panels 54A are each provided with two attachment flaps to secure the adjacent side and end wall panels thereto. It will be understood, however, that any suitable arrangement of the attachment flaps may be used to secure the support wall panels together to for the support wall 26. In the present arrangement, one or more of the wall panels 54 are provided with an attach-

ment flap 56 extending from a lateral edge of the wall panel 54 and arranged to contact an adjacent wall panel 54 during assembly of the support 14. It will be understood that the attachment flap(s) 56 will be attached to the adjacent wall panel 54 via the adhesive, which may be applied or coated on the attachment flap 56 and/or the adjacent wall panel 54.

[0074] As described above in connection with the support 14 of FIG. 4, the support flange 28 is provided as a series of first support flange parts 52 that form the support flange when the support 14 is assembled. In the embodiment shown, a first support flange part 52 extends from a distal edge of each of the end and side support wall panels 54.

[0075] In some embodiments, a second support flange part 52A extends from a distal edge of each of the corner wall panels 54A. The second support flange parts 52A provide support to the corner regions of the container flange 20 when the support flange 58 is attached to the container flange 20 under pressure. The second support flange parts 52 help to provide a uniform seal around the container flange and may increase the strength of the assembled sealed container assembly 10. The second support flange parts 52A may define a curved outer. The first support flange parts may define a substantially linear outer edge. The term “outer” here refers to the edge of the support flange parts 52, 52A furthest away from the mouth 30. In the arrangement shown, the support flange parts 52 are configured and arranged to form a continuous flange around the support wall 26. In alternative embodiments, the second support flange parts may be omitted. In such embodiments, the first support flange parts 52 may project beyond the respective end or side support wall panels 54 so as to define a continuous flange around the support wall 26. This arrangement reduces the number of breaks in the container flange, which has been found to increase the strength of the container flange, and of the assembled container assembly 10.

[0076] The container flange 20 includes a substantially flat or planar lower surface 42. The lower surface 42 defines an underside of the container flange 20. The support flange 28 is configured and arranged to abut against the lower surface 42 of the container flange 20, as is shown in FIG. 3. Put another way, the container flange is seated on the support flange. The support flange 28 is attached to the lower surface 42 of the container flange 20 by a second heat sealing material (not shown). The second heat sealing material may be a non-plastic heat sealable material, such as polylactic acid. In some arrangements, the second heat sealing material may be an aqueous heat sealing material, for example aqueous dispersions of low-density polyethylene and/or ethylene vinyl acetate. In some arrangements, the second heat sealing material may be polymeric film such polyethylene, polypropylene, polyethylene terephthalate. In some arrangements, the second heat sealing material may be a metal foil such as aluminium foil. The second heat sealing material may be the same or different to the first heat sealing material. It will be appreciated that the support flange 28 may be coated with the second heat sealing material, or the lower surface 42 of the container flange 20 may be coated with the second heat sealing material. The support 14, i.e. the support flange 28, is attached to the lower surface 42 by the application of heat and pressure.

[0077] The flexible container lid 36 and the support flange 28 are attached to opposing upper and lower surfaces 34, 42 of the container flange 20 via heat sealing materials. This arrangement enables the flexible container lid 36 and the

support flange 28 to be secured to the container flange 20 in a single step by applying heat and pressure to the upper and lower surfaces 34, 42 of the container flange 20.

[0078] Although not illustrated, the container assembly of Figure include a flexible container lid, as is discussed with reference to FIGS. 1 to 4.

[0079] Referring now to FIG. 7, the container flange 20 of the second embodiment includes first and second lengthwise portions 20A, 20B, which extend parallel to the length of the container 12; first and second width-wise portions 20C, 20D which extend parallel to the width of the container 12; and first, second, third and fourth corner portions 20E, 20F, 20G, 20H which extend between and connect the lengthwise and width-wise portions 20A, 20B, 20C, 20D, to form a continuous container flange 20. The mouth 24 of the container 12 defined by the container flange 20 has eight sides corresponding to the eight wall panels of the container wall 18. These include: first and second sides 62A, 62B which extend parallel to the length of the container 12; third and fourth sides 62C, 62D which extend parallel to the width of the container 12; a fifth side 62E which extends between and connects the first and the third side 62A, 62C; a sixth side 62F which extends between and connects the second and the fourth sides 62B, 62D; a seventh side 62G which extends between and connects the first and the fourth sides 62A, 62D; and an eight side 62H which extends between and connects the second and the third sides 62B.

[0080] It will be appreciated that the first and second sides 62A, 62B of the container mouth 24 are defined respectively by the first and second lengthwise portions of the container flange 20A, 20B; the third and fourth sides 62C, 62D of the container mouth 24 are defined respectively by the third and fourth width-wise portions of the container flange 20C, 20D; the fifth side 62E of the container mouth 24 is defined by the first corner portion of the container flange 20E; the sixth side 62F of the container mouth 24 is defined by the second corner portion of the container flange 20F; the seventh side 62G of the container mouth 24 is defined by the third corner portion of the container flange 20G; and the eighth side 62H of the container mouth 24 is defined by the fourth corner portion of the container flange 20H.

[0081] Containers 12 according to the second embodiment are designed to be supplied to food service facilities in a stacked configuration, and later de-nested and assembled with the supports 14 on site. To facilitate efficient de-nesting, the containers 12 are provided with features which interfere with corresponding features on adjacent containers 12 when the containers 12 are stacked in an alternating ABAB arrangement in which each container 20 is rotated 180 degrees relative to the adjacent containers 20 in the stack. Put another way, the containers 12 are provided with an asymmetric stacking formation. The asymmetric stacking formation may include a plurality of asymmetric recesses or depressions. In some embodiments, the container flange 20 is provided with depressions 60A-D which interfere when the containers 12 are stacked in an ABAB arrangement so as to increase a separation between adjacent containers 12 in the stack. It will be appreciated that in alternative embodiments, two, three, five or any suitable number of depressions may be provided. Providing a series of recesses and corresponding projections help to reduce the area of interference between the containers 12 when they are stacked together, which facilitates separation of the two containers, for example by an automatic destacking device.

[0082] The first, second, third and fourth depressions 60A,B,C,D are each positioned at a respective corner portion of the container flange 20E,20F,20G,20H. The first, second, third and fourth depressions 60A,B,C,D are contiguous with, and extend outwards from, the fifth, sixth, seventh and eighth sides 62E,62F,62G,62H of the container mouth 24, respectively. The depressions 60A,B,C,D extend through the thickness of the container flange 20 such that both the upper and lower surfaces of the container flange 20 at the depressions 60A,B,C,D is recessed relative to the rest of the container flange 20.

[0083] In order that the depressions 60A,B,C,D of adjacent containers 12 in the stack interfere, the positioning of the depressions 60A,60D on the first and fourth container flange corner portions 20A,20D differs from the positioning of the depressions 60B,C on the second and third container flange corner portions 20B,20C. That is, the positioning of the depressions 60A,D on the container flange corner portions adjacent the third width-wise portion of the container flange 20B differs from the positioning of the depressions 60B,60C on the container flange corner portions adjacent the fourth width-wise portion of the container flange 20C. Specifically, the first and fourth depressions 60A,60D are positioned towards “outer” ends of the fifth and eighth sides, respectively, where the term “outer” here refers to an end of the respective side which is relatively further away from a longitudinal centre line X of the container 12. On the other hand, the second and third depressions 60C,60E are positioned towards “inner” ends of the sixth and seventh sides 62F,62G, respectively, where the term “inner” here refers to an end of the respective side which is relatively further away from a longitudinal centre line X of the container 12. As a result, the depressions 60A,B,C,D of adjacent containers 12 in the stack do not coincide when the containers 12 are stacked in the alternating ABAB configuration. Instead, the depressed lower surfaces of the depressions 60A,B,C,D abut against the upper surface of the flange 20 of the container 12 directly below in the stack.

[0084] The container assembly 10 of the second embodiment may, if desired, be stacked in assembled form with the containers 12 inserted in the supports 14. Accordingly, stacks of pre-assembled container assemblies 10 may be provided to a food service facility, thus removing the need for the container assemblies 10 to be assembled on site. To facilitate stacking of the pre-assembled container assemblies 10, the base 16 of the container 12 of the second embodiment is provided with a number of base ribs 58. The base ribs 58 act to maintain the separation between the base 16 of any given container 20 and the base 32 of the support 14 of the adjacent container assembly 10 in the stack. This improves the ease of de-nesting separate container assemblies 10. The base ribs 58 also provide the added benefit of supporting solid food items above any juices which may accumulate in the bottom of the container during transport and storage of the packaged food items, thus helping to keep food fresh.

[0085] Although the teachings have been described above with reference to one or more preferred embodiments, it will be appreciated that various changes or modifications may be made without departing from the scope as defined in the appended claims.

1. A container assembly comprising:

a plastic container comprising a base, and a container wall extending from and surrounding the base and terminat-

ing at a container flange extending around the container wall to define a mouth of the container;

a support comprising a support wall arranged to surround the container wall, said support wall terminating at a support flange; and

a flexible container lid attached to an upper surface of the container flange by a first heat sealing material,

wherein the support flange is configured and arranged to abut against a lower surface of the container flange, and

wherein the support flange is attached to the lower surface of the container flange by a second heat sealing material.

2. The container assembly according to claim 1, wherein the container flange comprises a downwardly projecting lip extending from a peripheral edge of the container flange in a direction towards the base, and wherein the lip opposes the container wall to define a channel therebetween receiving the support flange therein.

3. The container assembly according to claim 2, wherein the support flange is mechanically secured between the lip and the container wall.

4. (canceled)

5. The container assembly according to claim 1, wherein the support comprises a tab projecting from a distal edge of the support flange so as to extend beyond a distal end of the lip.

6. The container assembly according to claim 1, wherein the container wall comprises a plurality of container wall panels defining corner regions at an intersection between adjacent container wall panels and the base, and wherein each corner region is tapered or chamfered.

7. The container assembly according to claim 1, wherein the container wall comprises eight wall panels.

8. The container assembly according to claim 1, wherein the container wall comprises a plurality of container wall panels including a first pair of opposing side wall panels, a second pair of opposing end wall panels, and four corner container wall panels, each corner container wall panel extending between a side wall panel and an adjacent end wall panel, optionally wherein container wall panels define a substantially octagonal container.

9. The container assembly according to claim 7, wherein the container flange defines a perimeter that is substantially rectangular; optionally, wherein the perimeter defines a rectangle with curved corners.

10. The container assembly according to claim 1, wherein the support wall comprises a plurality of support wall panels, and wherein the support flange comprises a support flange part extending from a distal edge of each support wall panel.

11. The container assembly according to claim 10, wherein the support flange parts are configured and arranged such that they form a continuous support flange around the support wall.

12. The container assembly according to claim 1, wherein the support wall comprises a plurality of support wall panels including a first pair of side support wall panels, a second pair of end support wall panels, and four corner support wall panels each corner support wall panels extending a support side wall panel and an adjacent support end wall panel; optionally wherein the support wall panels define a substantially octagonal support.

13. The container assembly according to claim 1, wherein the support wall comprises eight support wall panels.

14. The container assembly according to claim **12**, wherein the support flange defines a perimeter that is substantially rectangular; optionally, wherein the perimeter defines a rectangle with curved corners.

15. The container assembly according to claim **1**, wherein the container wall is substantially linear in a direction between the base and the container flange and tapers in a direction towards the base.

16. The container assembly according to claim **15**, wherein the container wall tapers at an angle in the range 5° to 19° , in the range 10° to 18° , in the range 12° to 16° , for example approximately 14° .

17. (canceled)

18. (canceled)

19. (canceled)

20. The container assembly according to claim **1**, wherein the first heat sealing material is selected from a polymeric film such as polyethylene, polypropylene, and/or polyethylene terephthalate.

21. The container assembly according to claim **1**, wherein the second heat sealing material is selected from: a non-plastic heat sealable material such as polylactic acid; an aqueous heat-sealing material; polyethylene; polypropylene; and/or polyethylene terephthalate.

22. The container assembly according to claim **1**, wherein the support is formed from a fibrous material such as paper or card.

23. (canceled)

24. The container assembly according to claim **1**, wherein the support wall comprises a pair of support end wall panels and a pair of support side wall panels and a support corner panel between adjacent support side and end wall panels, and wherein the support flange parts extend from the support side and project beyond the respective end or side support wall panel so as to define a continuous flange around the support wall.

25. The container assembly according to claim **1**, wherein the melting temperature of the second heat sealing material is less than the melting temperature of the first heat sealing material.

26. (canceled)

27. (canceled)

28. (canceled)

29. (canceled)

30. (canceled)

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